

Designing Fractal Expressions for Immersive Aesthetic Experiences

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Figure 1: Interactive Video Presentation of Application Scenarios Developed with the persona.fractalis.II Framework

ABSTRACT

Aesthetic experiences are closely related to human emotions and experiences as well as the perception of aesthetic objects and phenomena. We consider interaction with generated immersive worlds to be an immersive aesthetic experience if it is designed independently of any functional use and can be experienced in an aesthetically pleasing way. The possibilities of generative techniques such as AI and other algorithms and visualization through the integration of virtual reality open up new aesthetic possibilities. We present the results of a collaboration between scientists and designers/artists in the development of a framework for three-dimensional fractals [5] and its application in various scenarios in the field of festivals, performances and exhibitions.

Index terms: Immersive aesthetics, 3D real-time fractals, VR

1 INTRODUCTION

We focus briefly on three key aspects of aesthetic immersion: Long-standing research on aesthetic perception since the eighteenth century argues that aesthetic experience does not only arise from the contemplation of artistic objects and representations, but is also part of artistic practice, i.e. the active production of such objects and phenomena [1]. Aesthetic experience is associated not only with the reception and production of artistic artefacts, but also with the perception of alternative aesthetic entities, e.g. from nature, science, society, or technology. Marković [2] argues that there is a broad consensus that aesthetic experience is a state of mind that is

qualitatively different from everyday experience and is considered to be a psychological process in which attention is focused on one entity and all other entities are relegated to the background.

As VR technology evolves, it increasingly uses aesthetics to create immersive environments that provide unique experiences that are often unattainable in traditional media. Furthermore, the interactivity and immersion inherent in VR distinguishes it from other digital technologies. Zhu and Li discuss the concept of "aesthetic interaction", which focuses on how narrative and visual elements can engage users more deeply than traditional human-computer interactions, which often prioritise performance over engagement. The design of VR environments plays an important role in shaping aesthetic experiences. Boz et al. show that architectural variables, such as room shape and texture, significantly influence users' aesthetic evaluations, suggesting that careful design can enhance the overall immersive experience [3]. Similarly, Rani emphasises the need for a multidimensional evaluation approach to assess both immersive and aesthetic experiences in VR [4].

If we consider the development and presentation of digital content in many areas of social life, our research perspective can be summarised as follows. (1) Immersive technologies have a special potential for aesthetic experience, as immersion and interaction enable a strong presence of the recipient in the virtual world. (2) Through real-time interaction during the experience, the recipient is positioned between states of aesthetic reception and production, which in our view is a further supporting factor. (3) Since the modification and exploration within the immersive fractal world takes place in real time based on user interaction and, in contrast to classical 3D user interface concepts, is detached from a concrete user task, there is an additional experience-based focus on the interaction with the fractals.

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2 APPLICATION SCENARIOS FOR IMMERSIVE FRACTALS

The persona.fractalis.II framework has been used to develop distinctive aesthetic experiences for a variety of exhibition and festival projects. A key aspect of this development has been the facilitation of seamless access for diverse audiences, together with the comprehensive realization of features such as immersion, interaction, and ease of use. An overview of selected 3D experts within our team is provided in table 1.

Project (*shown in https://is.gd/R6WJlh)	Immersion	Interaction	Ease of Use
1) LED screen*	+	0	+++
2) 360° View	++	+	++
3) VR-Viewscope	++	++	+
4) VR HMD + Ctrl*	+++	+++	0
5) Fractal VJ*	++	++	++
6) Invisible Cities*	++	++	+
7) Fractal Flight	+++	++	+
8) Fractal Dome*	+++	0	+++
9) Fractal Theremin*	++	++	+++

Table 1. Expert Review of persona.fractalis.II Projects

The LED wall project "epic.fail" (1) is a curated flight through a series of interconnected fractal worlds created by an experienced 3D artist, an epic music composer and a video expert. The results were shown at a festival on a 7m x 7m LED wall, providing very easy access to a large presentation area, but without any interaction. Interactive 360° video versions (2) of some of the curated fly-throughs were created and can be experienced using a Youtube 360° viewer. Curated fly-throughs can be saved and experienced in real time. For a digital art festival, we used this variant together with a home-made VR telescope device (3) to provide easy access to an immersive and interactive experience. This provides a more intuitive but less interactive access than a VR HMD with controller (4). The control of a fractal flight with full body, wind and scent output (7) was presented in a project with an ICAROS device (see Figure 3, right).



Figure 2: VR-Viewscope and Fractal Dome Experience

Another user-friendly but less interactive variant was the production of a video tour for an immersive dome location in Wuppertal, Germany (8). The 47-meter-high projection, equipped with 33 beamers and 60 million pixels, is among the most substantial 360° screens in Europe. A semi-interactive approach was utilized in a techno-theatre piece entitled "State of Euphoria" (5), which aimed to provide an accessible experience for a broader audience. The system incorporated video clips of diverse fractal environments, referred to as biomes, which were generated and loaded into the system. The video segments can be modified in real-time using digital post-processing effects, which are applied through an OSC/MIDI-based control board by the VJ (see figure 3, left). The OSC-interface developed for this system was also employed to establish a real-time connection with a theremin, an electronic instrument for touchless play (9). The "Invisible Cities" project further discusses the idea of automated aesthetics. Here, the main content that the user interactively explores is computer generated. Based on texts from Italo Calvino's "The Invisible

Cities", we generated new texts describing cities, automatically translated them into several languages and generated output using a Text2Speech engine. The fractal worlds serve as illustrations of the generated lyrical content.

3 TECHNICAL IMPLEMENTATION

Our system is a software prototype that generates three-dimensional fractals in real time and makes them interactively explorable. In the actual prototype instances of our 3D fractal system, we have parameterised different versions of the Mandelbox fractal and other well-known fractal types with different colour schemes and displayed them in real time in virtual reality in our self-developed 3D system (based on Unity3D) and interactively animated the movement through the fractal world. Navigation along a predefined path is possible, as is saving, retrieving and interpolating between different parameter sets. As the user explores the fractal world, the parameters of the fractal world can be interactively modified through various interfaces to input devices (sensors, controllers, midi devices). This allows a wide range of aesthetic experiences to be realised and allows artists to focus on the design of these experiences. Pre-produced explorations are also possible via a video-based component that can be influenced in real time by interactive elements (e.g. controllers). This option limits the scope of design, but allows easier navigation through the complex fractal worlds and provides a choreographed experience. The use of AI systems and algorithms offers automated content creation for spoken text, music and sound effects, allowing designers and artists to engage with synthetically generated content in combination with their own means of expression.

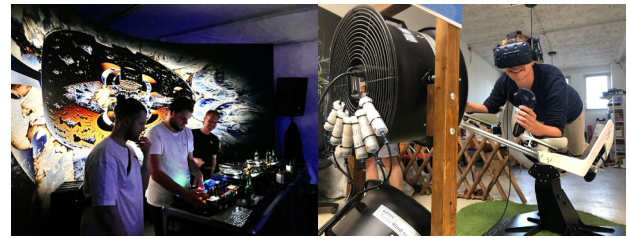


Figure 3: Fractal VJ and Fractal Flight

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